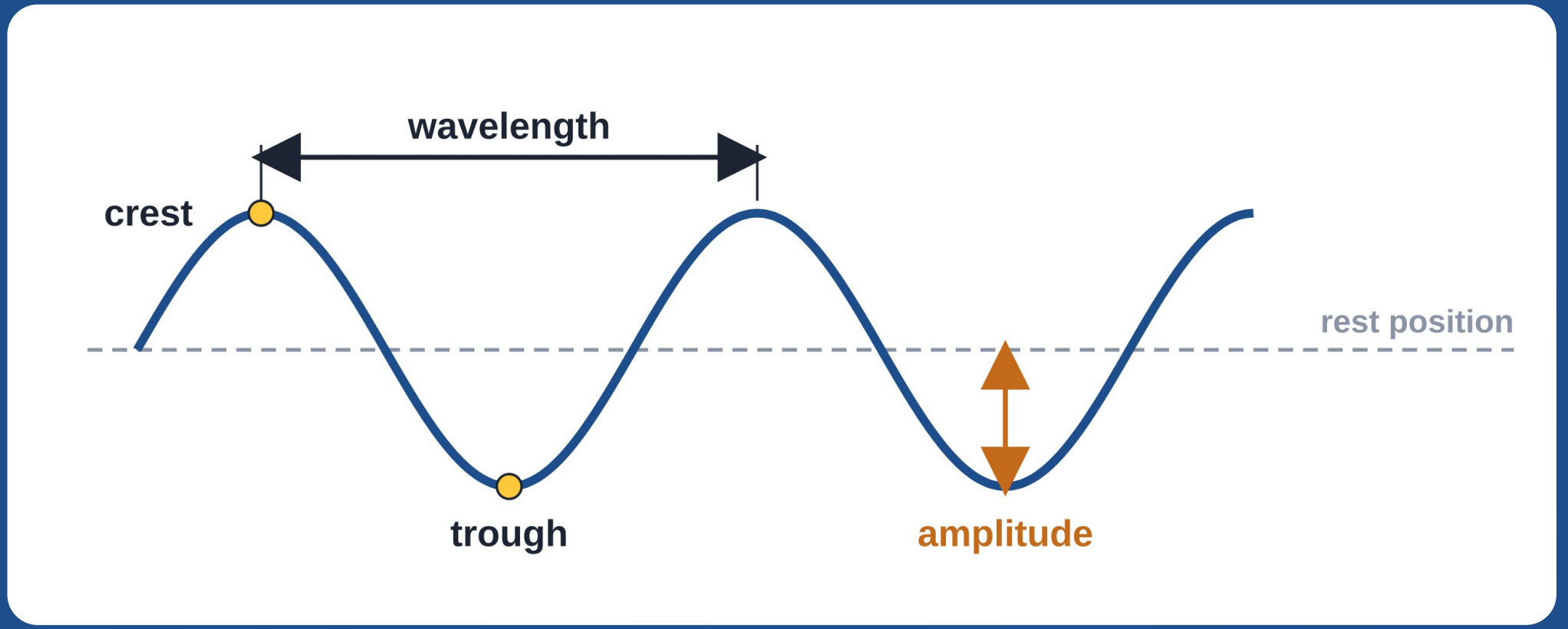
4.01 Describe a Wave

What is a wave, and how do we measure one?



Today's Goals

I can explain that a wave carries energy from one place to another.

I can tell a transverse wave from a longitudinal wave.

I can label a crest, trough, rest position, wavelength, and amplitude.

I can measure wavelength and describe frequency.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Waves can be described by their speed, wavelength, amplitude, and frequency.
- Vibrations cause wave-like disturbances that carry energy.
- Mechanical waves need a medium; the medium returns to its rest position.
- Waves can be transverse or longitudinal.

Inquiry skills (Grades 5–8): "Develop descriptions, models, explanations and predictions" and "Think critically and logically to connect evidence and explanations" (today's CER).

In this lesson

- What a wave is
- Transverse vs. longitudinal
- Parts of a wave
- Measuring wavelength and amplitude
- Frequency and speed
- Graph CER

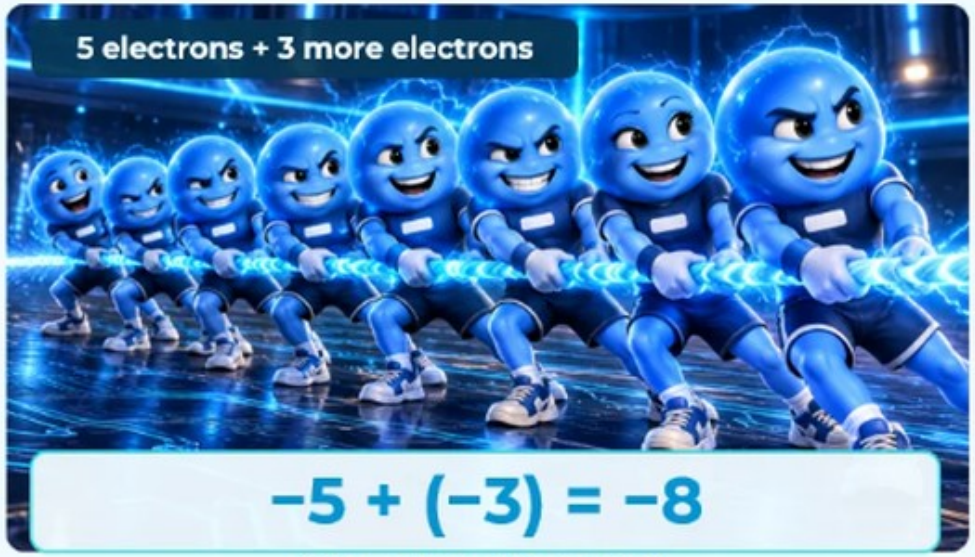
Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

BATTLE OF THE CHARGES

Adding integers with electrons and protons

SAME SIGN? TEAM UP!

Add the counts. Keep the sign.



$-5 + (-3) = -8$

Same signs: add and stay!

OPPOSITE SIGNS? PAIR UP!

Subtract. The sign with more wins.



$-7 + 3 = -4$

3 pairs total 0. Four extra electrons!

ELECTRON = −1 PROTON = +1

"More" means greater absolute value. Pairs sum to 0; particles do not disappear.

CASE #2 • TUESDAY

Is the buoy at its rest position?

WAVE WORD: *Amplitude*
The distance from the rest position to a crest or trough.

MYSTERY

A buoy is 2 m below the rest position. It moves up 2 m. Is the buoy at its rest position?

CLUES

- Start: −2 m
- Move up: +2 m
- Rest: 0 m

BUOY HEIGHT (m)

+3
+2 — Crest +2 m
+1
0 — Rest 0 m
−1
−2 — Trough −2 m
−3



CLAIM

Answer the mystery question.

Is the buoy at its rest position?

☐ YES ☐ NO

EVIDENCE

Show your work!

Solve:

−2 + 2 = _____

REASONING

Explain the rule.

Finish the sentence:

"−2 and 2 are opposites, so..."

Vocabulary

wave

a disturbance that carries energy from one place to another

vibration

a quick back-and-forth or up-and-down motion that starts a wave

medium

the material a wave travels through (rope, water, air)

rest position

where the medium sits when no wave is passing

transverse

the medium moves up and down, across the wave

longitudinal

the medium moves back and forth, along the wave

crest / trough

the highest point / the lowest point of a wave

cycle

one complete wave that repeats (one crest + one trough)

wavelength

the length of one cycle, crest to crest

amplitude

the distance from the rest position to a crest

frequency

how many cycles pass in a period of time

speed

how far the wave travels each second

Waves Song

Listen for these words:

- energy
- medium
- crest and trough
- wavelength and amplitude
- frequency
- transverse and longitudinal

Type one word you heard in the chat!



Flocabulary: Waves (plays right here in Slide Show)

▶️ **Backup: open on YouTube**

What Is a Wave?

A wave carries energy

A wave is a disturbance that carries energy from one place to another.

Vibrations cause waves

Pluck a guitar string and it vibrates. The vibration starts a wave.

The medium goes back

The medium (rope, water, air) moves, then returns to its rest position. Energy travels on.

Watch: guitar strings up close

Look for: Which way does each string move? What makes the sound?



Open the 4.01 page

Mechanical waves (rope, water, sound) need a medium to travel through.

Waves Are Everywhere

Ocean waves

Watch a floating object. Does the water travel to the shore, or does the energy?

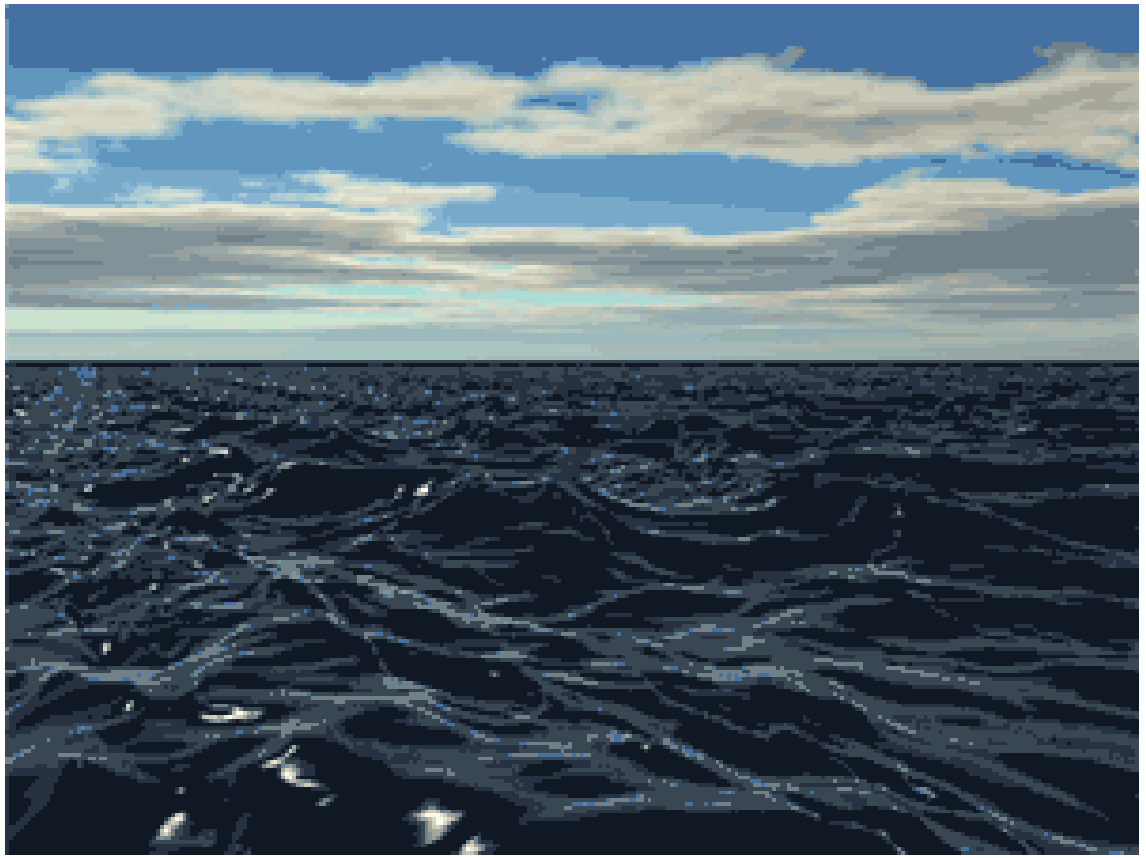
Rope waves

An athlete shakes exercise ropes. Which way does the rope move? Which way does the energy go?

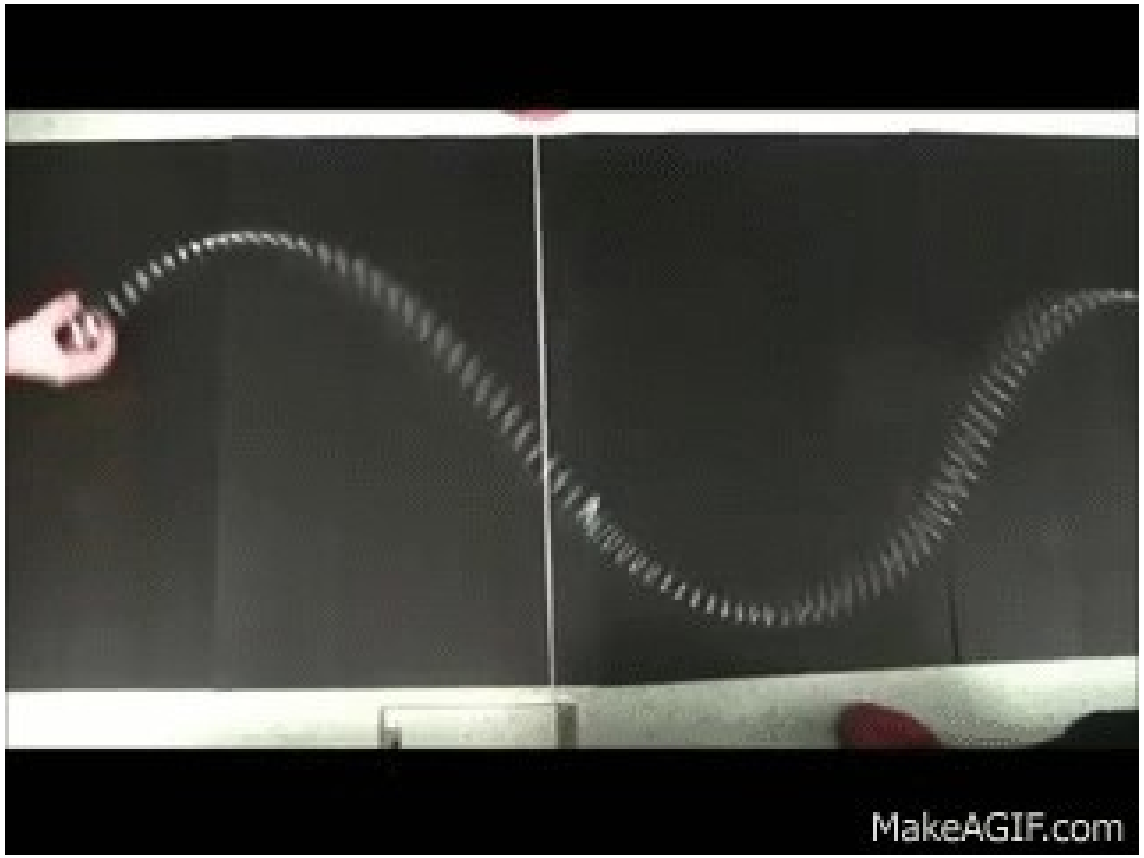
▶️ **Play the PBS videos (4.01 page)**

Extra videos: these are on the 4.01 page but not in the Classkick.

See It Move



Ocean: the water moves up and down in place. The energy moves across the ocean.



Spring toy: the coils move side to side, across the direction the energy travels: a transverse wave.

Engageli Poll

Classkick: 4.01 Part 1 · Q2

Guitar-string waves and rope waves both carry _____ from one place to another.

Single choice poll

A. the string

Single choice poll

B. energy

Single choice poll

C. air

Single choice poll

D. the rope

Fill the Gaps

Waves are caused by _____. A mechanical wave carries _____ from one place to another. It needs a _____ to travel through. The medium moves, then goes back to its _____ position.

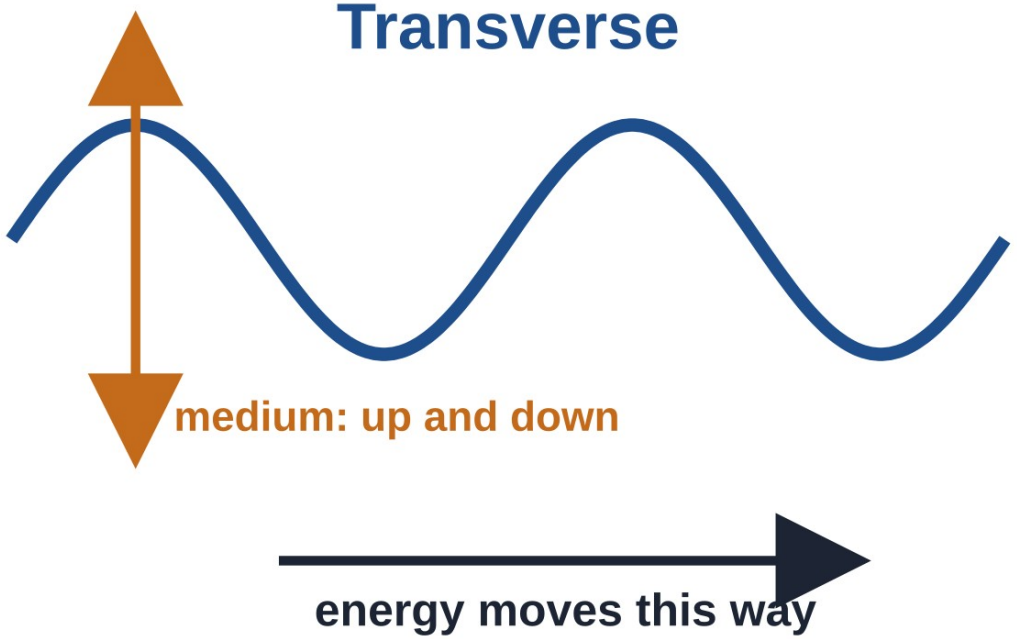
Word bank:

medium	**speed**	**rest**	**vibrations**	**energy**

One extra word.

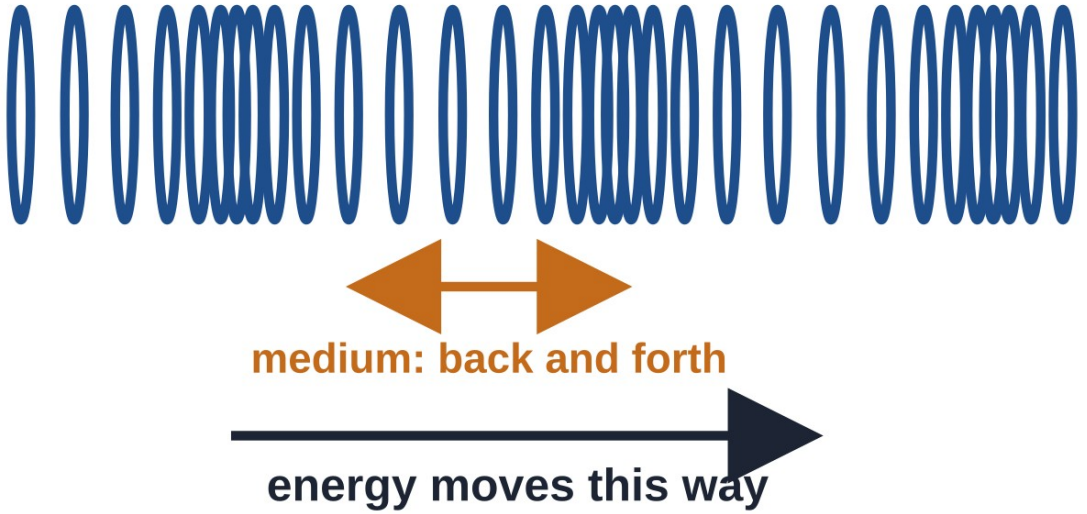
Two Ways a Medium Can Move

Transverse



Transverse: the medium moves up and down, across the direction the energy travels. Example: shaking a rope.

Longitudinal



Longitudinal: the medium moves back and forth, along the direction the energy travels. Example: pushing a spring toy.

Engageli Poll

Classkick: 4.01 Part 1 · Q3

You push and pull one end of a spring toy. The coils move back and forth, the same way the energy travels. What kind of wave is it?

Single choice poll

A. transverse

Single choice poll

B. longitudinal

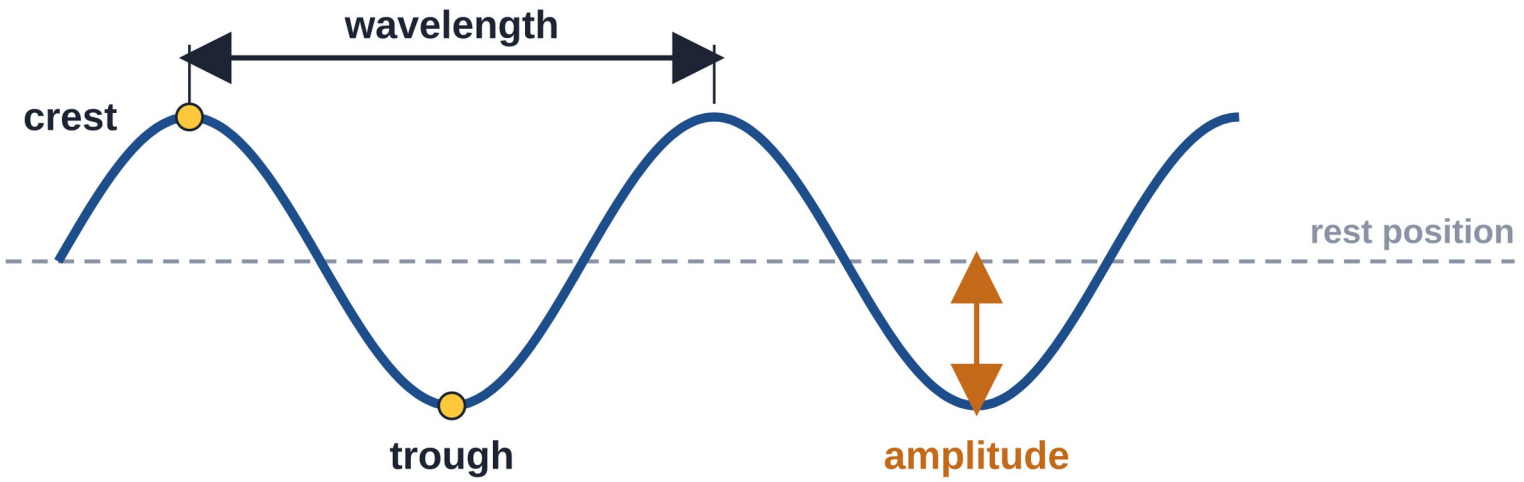
Single choice poll

C. not a wave

Single choice poll

D. both kinds

Parts of a Wave



Crest: high point

Trough: low point

Rest position: the flat middle line

Wavelength: one full cycle

Amplitude: rest position to a crest or trough

▶️ Practice: Label the Parts (Wordwall)

Measuring a Wave

Cycle

One complete wave that repeats: one crest + one trough. A waveform is a curve that shows the wave's shape.

Wavelength

Measure crest to crest OR trough to trough. You get the same answer both ways!

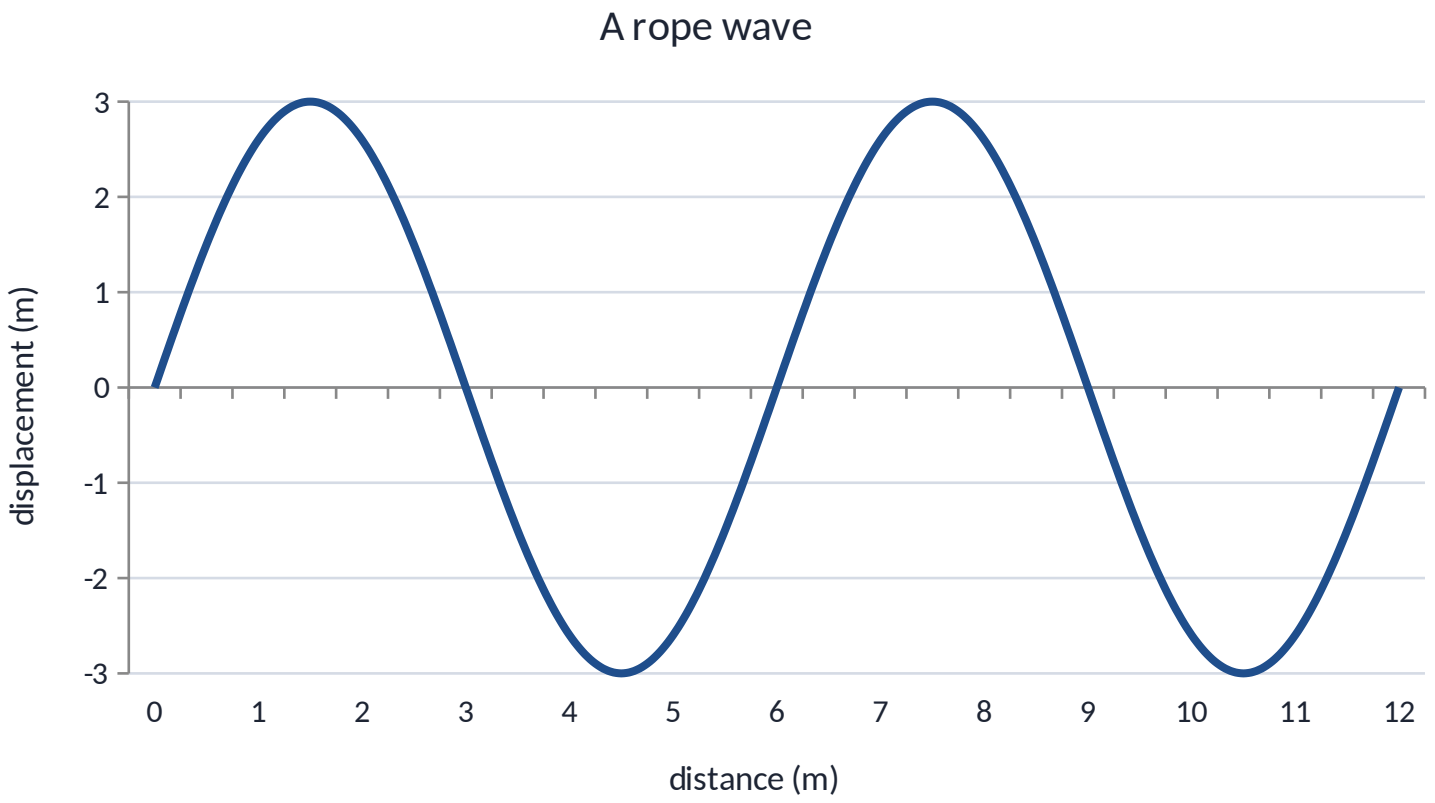
Amplitude

Measure from the rest position to a crest (or to a trough). Bigger amplitude = more energy.

▶️ **Watch: Measuring Waves animation**

Classkick: 4.01 Part 2 · Q5

Measure It Two Ways



Crest to crest: _____ m

Trough to trough: _____ m

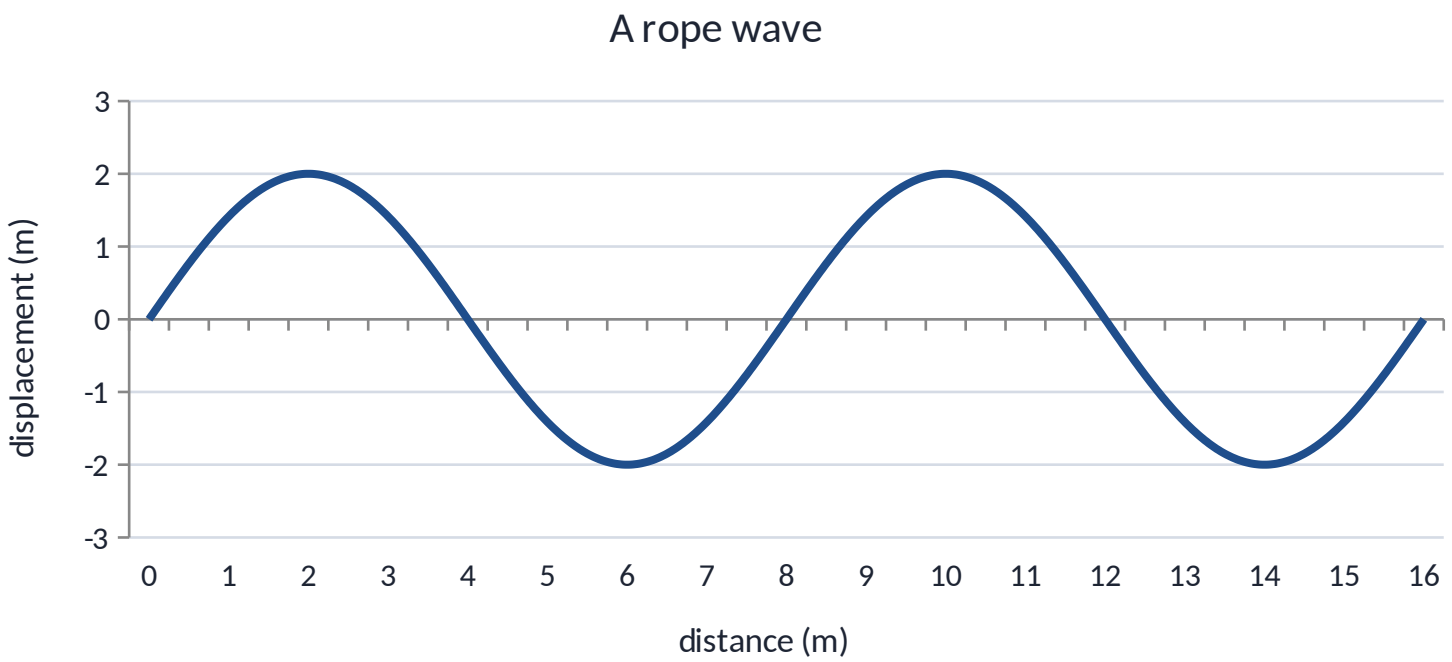
Same both ways? YES / NO

Amplitude: _____ m

We do this one together. You do the rope wave on your Classkick (Part 2, Q5).

Classkick: 4.01 Part 2 · Q5

Graph Poll



Read the graph.

What is the amplitude of this wave?

Hint: start at the rest position (0).

Single choice poll

A. 2 m

Single choice poll

B. 4 m

Single choice poll

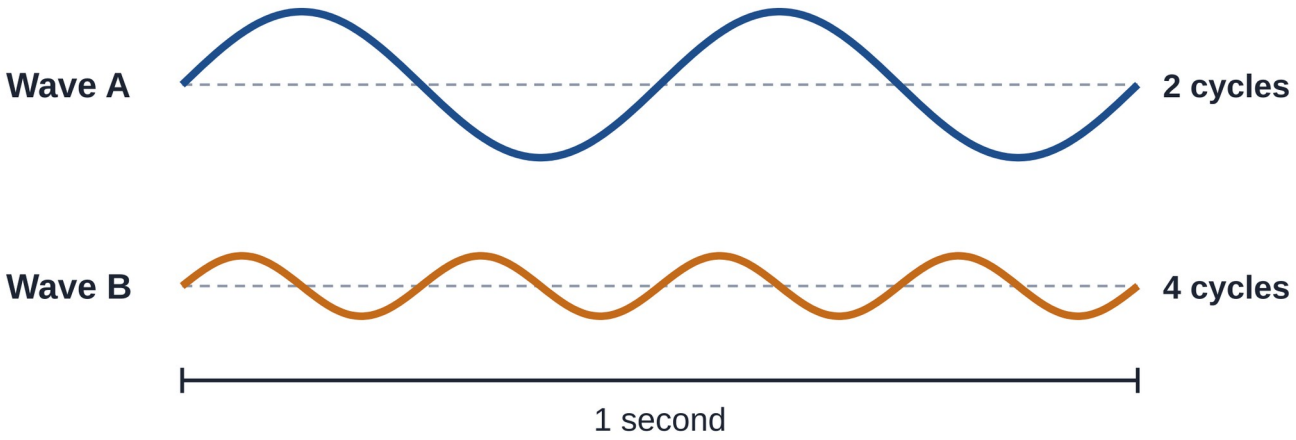
C. 8 m

Single choice poll

D. 16 m

Classkick: 4.01 Part 2 · Q6–Q7

Frequency and Speed



Frequency = how many cycles pass in a period of time. Wave B has the higher frequency. Wave A is taller (bigger amplitude), so it carries **more energy**.

Speed

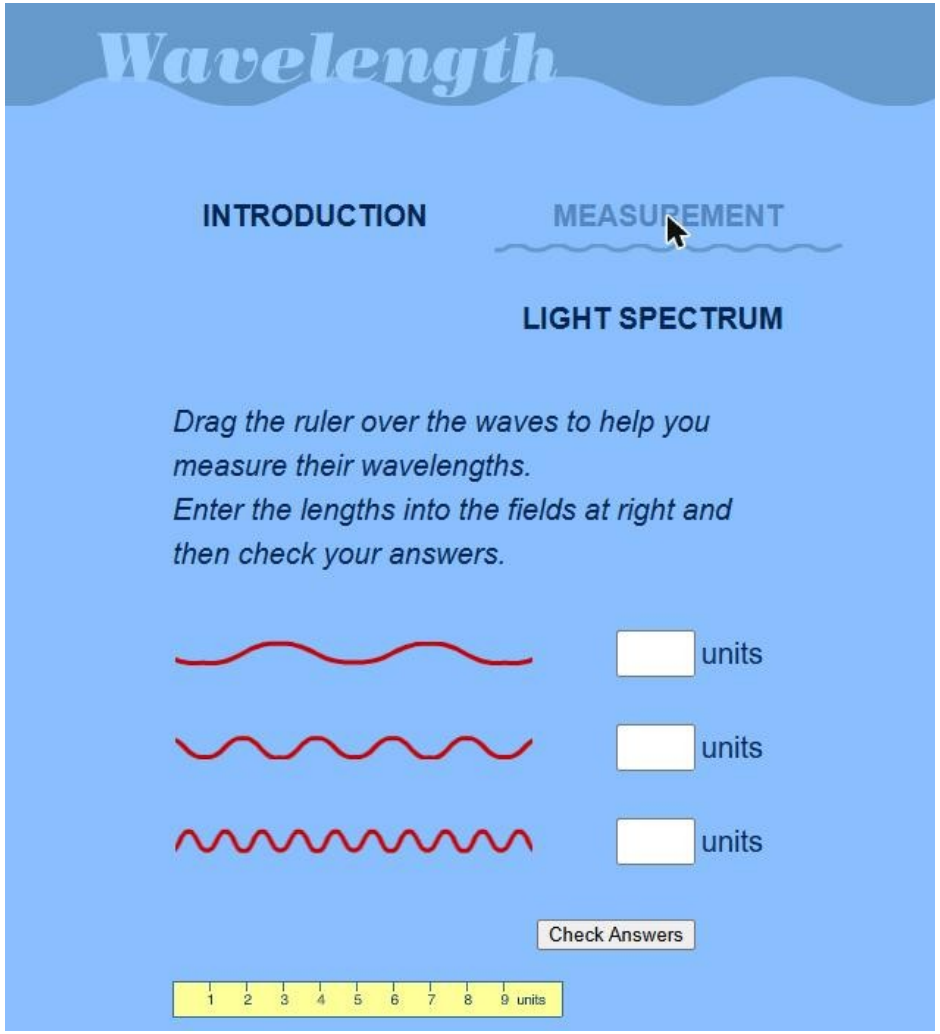
speed = wavelength × frequency

A wave is 4 m long and 3 waves pass each second.
4 m × 3 per second = **12 m/s**

Light is a wave too. Red light has a longer wavelength than violet light.

▶ PBS: Measuring Wavelengths

PBS: Measuring Wavelengths



Click **Measurement**.

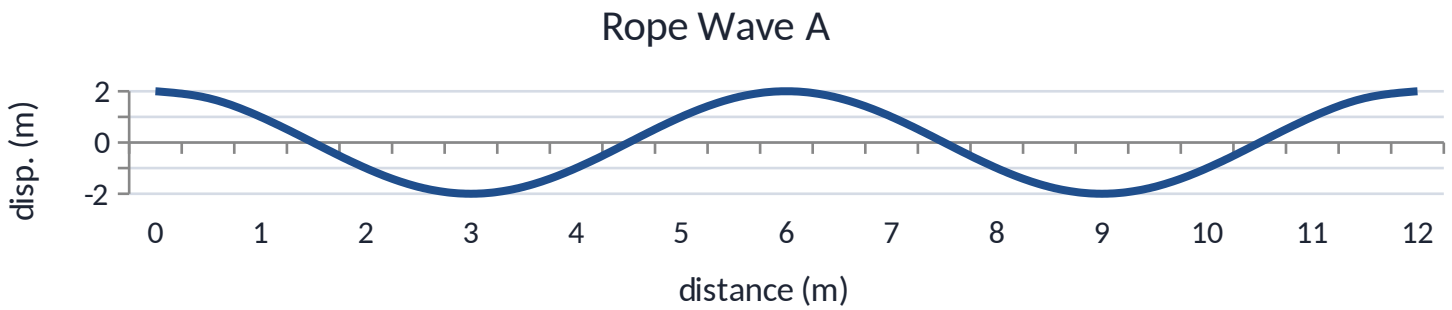
Drag the yellow ruler onto a wave. Measure crest to crest.

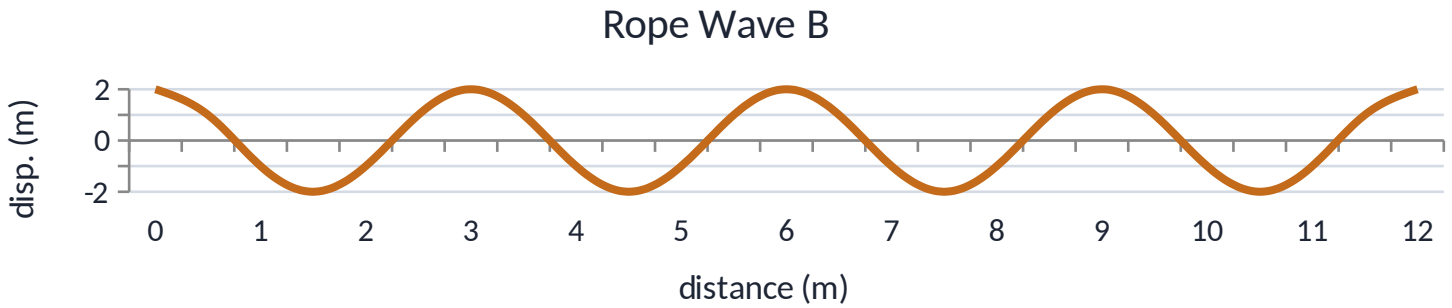
Type each wavelength, then click **Check Answers**.

Then click **Light Spectrum**: which has the longer wavelength, red or violet?

▶️ **Open the PBS interactive**

CER 1: Claim





Question

Which rope wave has the longer wavelength?

Claim: pick one.

Single choice poll

A. Wave A

Single choice poll

B. Wave B

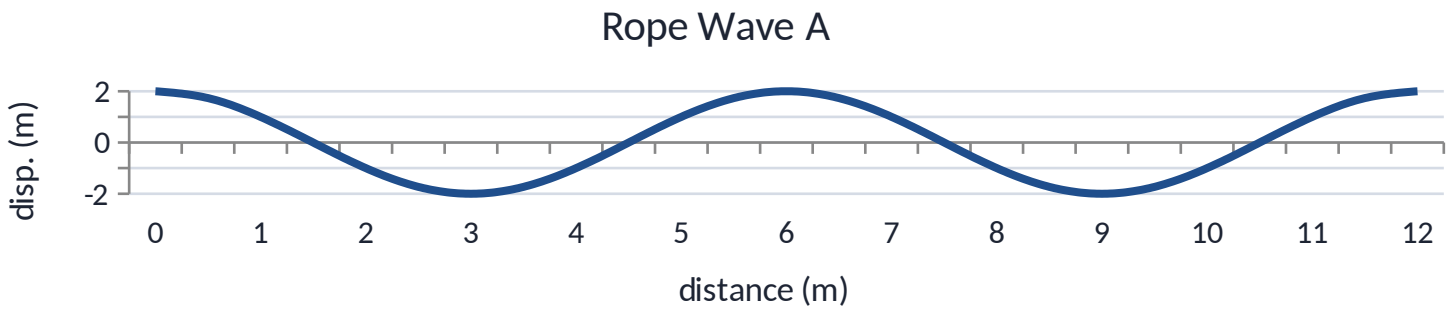
Single choice poll

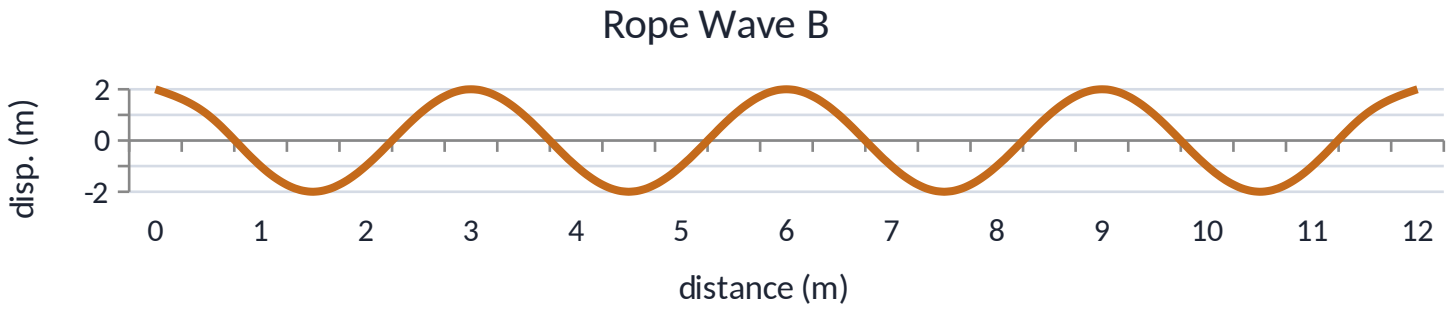
C. They are the same

Single choice poll

D. Can't tell from a graph

CER 2: Evidence





Fill in the blanks

Type your numbers in the chat.

Wave A has crests at 0 m, **(1)** ____ m, and 12 m.

Wave A's wavelength is **(2)** ____ m.

Wave B's wavelength is **(3)** ____ m.

Evidence frame: On the graph, Wave A's crests are _____ m apart and Wave B's crests are _____ m apart.

CER 3: Reasoning

Reasoning: finish the sentence.

Wavelength is the distance from _____. Wave A's crests are farther apart, so Wave A has the longer wavelength.

Single choice poll

A. one crest to the next crest

Single choice poll

B. a crest to a trough

Single choice poll

C. the rest position to a crest

Single choice poll

D. the top of the graph to the bottom

Key Ideas

1. A wave is a disturbance that carries energy, not matter.
2. Vibrations cause waves. Mechanical waves need a medium, which returns to its rest position.
3. Transverse: up and down. Longitudinal: back and forth.
4. Wavelength: crest to crest. Amplitude: rest to crest. Frequency: cycles per time.

Re-rate your goals: 1 = Not yet, 2 = Getting there, 3 = I can teach it